

PRE-HARDEN & TEMPERED ALLOY STEEL

ALLOY STEEL KS D3707 D3711, JIS G4104 G4105

KS JIS	AISI	Chemical Composition %							Mechanical Properties					
		C	Si	Mn	P	S	Cr	Mo	YP	TS	EL	RA	Impact	Hardness
									kg/mm ²	kg/mm ²	%	%	charpy	HB
SCR415		.13/.18	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	-	≥80	≥15	≥40	≥6	217/302
SCR420	5120	.18/.23	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	-	≥85	≥14	≥35	≥5	235/321
		.17/.22	.15/.30	.70/.90	≤.035	≤.040	.70/.90							
SCR430	5130	.28/.33	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥65	≥80	≥18	≥55	≥9	229/285
		.28/.33	.15/.30	.70/.90	≤.035	≤.040	.80/1.10							
SCR435	5135	.33/.38	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥75	≥90	≥15	≥50	≥7	255/311
		.33/.38	.15/.30	.60/.80	≤.035	≤.030	.80/1.05							
SCR440	5140	.38/.43	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥85	≥100	≥12	≥40	≥5	285/341
		.38/.43	.15/.30	.70/.90	≤.035	≤.040	.70/.90							
SCR445	5147	.43/.48	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥85	≥100	≥12	≥40	≥5	285/341
		.46/.51	.15/.30	.70/.95	≤.035	≤.030	.85/1.15							
SCM415		.13/.18	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	-	≥85	≥16	≥40	≥7	235/321
SCM418	4118	.16/.21	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	-	≥90	≥15	≥40	≥7	248/331
		.18/.23	.15/.30	.70/.90	≤.035	≤.040	.40/.60	.08/.15						
SCM420		.18/.23	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	-	≥95	≥14	≥35	≥6	285/363
SCM421		.17/.23	.15/.35	.70/1.00	≤.030	≤.030	.90/1.20	.15/.30	-	≥100	≥14	≥35	≥6	285/363
SCM822		.20/.25	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.35/.45	-	≥105	≥12	≥30	≥6	302/415
SCM430	4130	.28/.33	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥70	≥85	≥18	≥55	≥11	241/293
		.28/.33	.15/.30	.40/.60	≤.035	≤.040	.80/1.10	.15/.25						
SCM432	4135	.27/.37	.15/.35	.30/.60	≤.030	≤.030	1.00/1.50	.15/.30	≥75	≥90	≥16	≥50	≥9	225/321
		.33/.38	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥80	≥95	≥15	≥50	≥8	269/321
SCM435	4140	.33/.38	.20/.35	.70/.90	≤.035	≤.040	.80/1.10	.15/.25						
		.38/.43	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥85	≥100	≥12	≥45	≥6	285/341
SCM440	4145	.38/.43	.15/.30	.75/1.00	≤.035	≤.040	.80/1.10	.15/.25						
		.43/.48	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥90	≥105	≥12	≥40	≥4	302/363
SCM445		.43/.48	.15/.30	.75/1.00	≤.035	≤.040	.80/1.10	.15/.25						